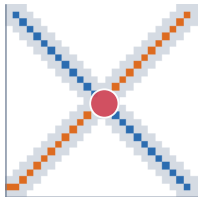


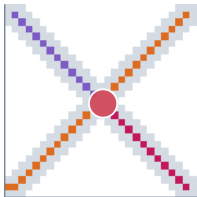
(a) matches the reference



$$F_1 = 1.000$$

P 1.00 R 1.00

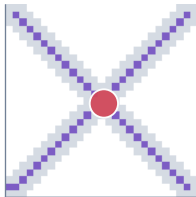
(b) one strand not linked



$$F_1 = 0.667$$

P 1.00 R 0.50

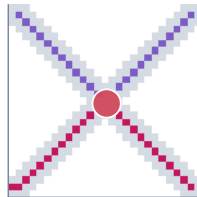
(c) both strands merged



$$F_1 = 0.500$$

P 0.33 R 1.00

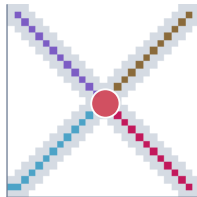
(d) arms mis-paired



$$F_1 = 0.000$$

P 0.00 R 0.00

(e) every arm separate



$$F_1 \text{ undefined}$$

P undefined R 0.00

Two strands cross. The scorer deletes the junction pixels (red) and keeps the four arms they separate, then scores the 6 pairs of arms: a pair is right when the grouping joins it, or leaves it apart, as the reference does. The reference is the pairing in (a); P and R are the precision and recall of those 6 pair decisions.